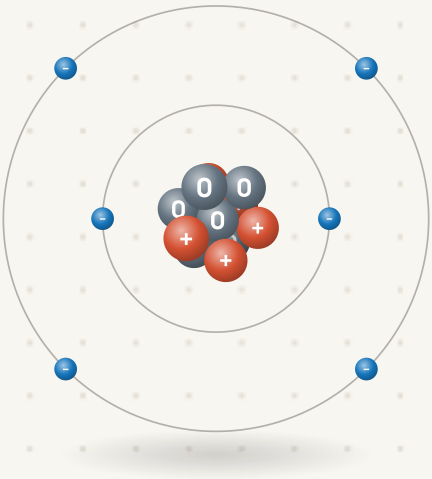
YEAR 11 PHYSICS, AQA GCSE, PART 4 OF 4

Atoms & Nuclear Radiation.

Part 4 of 4 Lessons 8–10

IN THIS BOOKLET

- Lesson 8: Half-life and random decay
- Lesson 9: Half-life calculations
- Lesson 10: Contamination and irradiation



NAME _____ CLASS _____

TEACHER _____ TARGET GRADE _____

START HERE

How this booklet works.

This is the booklet for **Lessons 8–10** of the Atoms and Nuclear Radiation unit. It builds on the lessons before it; the "Do now" questions and the quick answers at the back help you check what you remember. Keep it: it is your classwork book *and* your revision guide.

DO NOW
Quick questions from earlier lessons. Remembering again is what makes it stick.

LEARN IT
Explanations and diagrams. Fill in the slide examples with your teacher, then try "Your turn" alone.

TRY IT
Practice that builds up, then a challenge. Quick checks from the slides follow some lessons.

LAB
Record what you find in a simulation or a model experiment.

EXAM CORNER
AQA-style questions with a tip on how the marks are given.

REVISE IT
Summary, cover-and-answer questions and an "I can" check to colour in.

! **Revising at home?** (1) Read the Learn it pages. (2) Cover them and answer the "Cover the page" questions in Revise it. (3) Check with the quick answers at the back. (4) Go through the matching slides in the teaching deck, and try the simulations.

The route through the unit (this booklet highlighted)

01 Inside the atom

02 Isotopes and ions

03 From plum pudding to the nucleus

04 Energy levels and the nucleus

05 Radioactive decay

06 Nuclear equations

07 Properties and uses of radiation

08 Half-life and random decay
Slides 35–37

09 Half-life calculations
Slides 36

10 Contamination and irradiation
Slides 38–41

Teaching deck: **panphy.app/gcsephy/decks/atoms-and-radiation**. Simulations: **panphy.app/simulations/atomic_models.html** and **panphy.app/simulations/nuclear_decay.html**

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LESSON

08

Half-life and random decay.

AQA Physics 4.4.2.3; Combined Science 6.4.2.3

BIG QUESTION

You cannot predict when one nucleus will decay. So how can we say that half of a sample of iodine-131 will be left after 8 days?

BY THE END YOU CAN

- explain why decay is random but a large sample is predictable
- define half-life in two ways
- find the half-life from a decay curve
- plot a decay curve from data

half-life | random | decay curve | sample | count rate | activity

SLIDES AND SIMULATIONS

Teaching deck, slides 35–37

Half-life, using half-life and random decay.

panphy.app/gcsephy/decks/atoms-and-radiation/#35

Nuclear Decay simulation

panphy.app/simulations/nuclear_decay.html

RETRIEVE

Do now

From earlier lessons. No notes!

1 Which radiation is stopped by paper?

2 What does a count rate measure?

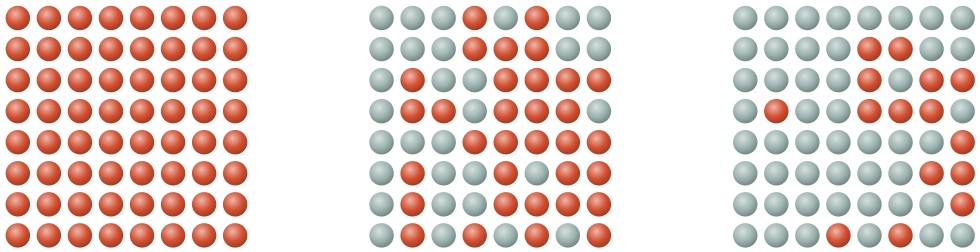
3 What is the unit of activity?

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EXPLAIN

Random, but predictable

Random, but predictable



start
64 undecayed

after 1 half-life
33 undecayed (≈ 32)

after 2 half-lives
15 undecayed (≈ 16)

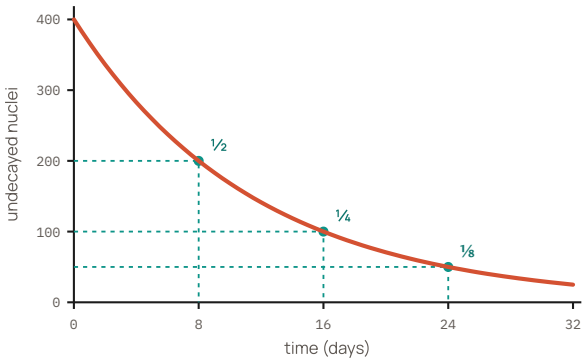
Every unstable nucleus of an isotope has the **same chance** of decaying in a given time. You cannot say which one will decay next, or when. But in a **large sample**, the number that decay in a given time is very predictable. In a small sample, the numbers vary more.

THINK OF COINS

Throw 100 coins and remove every head. About half are left. Throw the rest and remove the heads again: about a quarter are left. You cannot predict any one coin, but you can predict the group.

Where the model breaks: coins only 'decay' when you throw them, but nuclei decay at any moment; and real samples have billions of nuclei.

Half-life



The **half-life** of a radioactive isotope is:

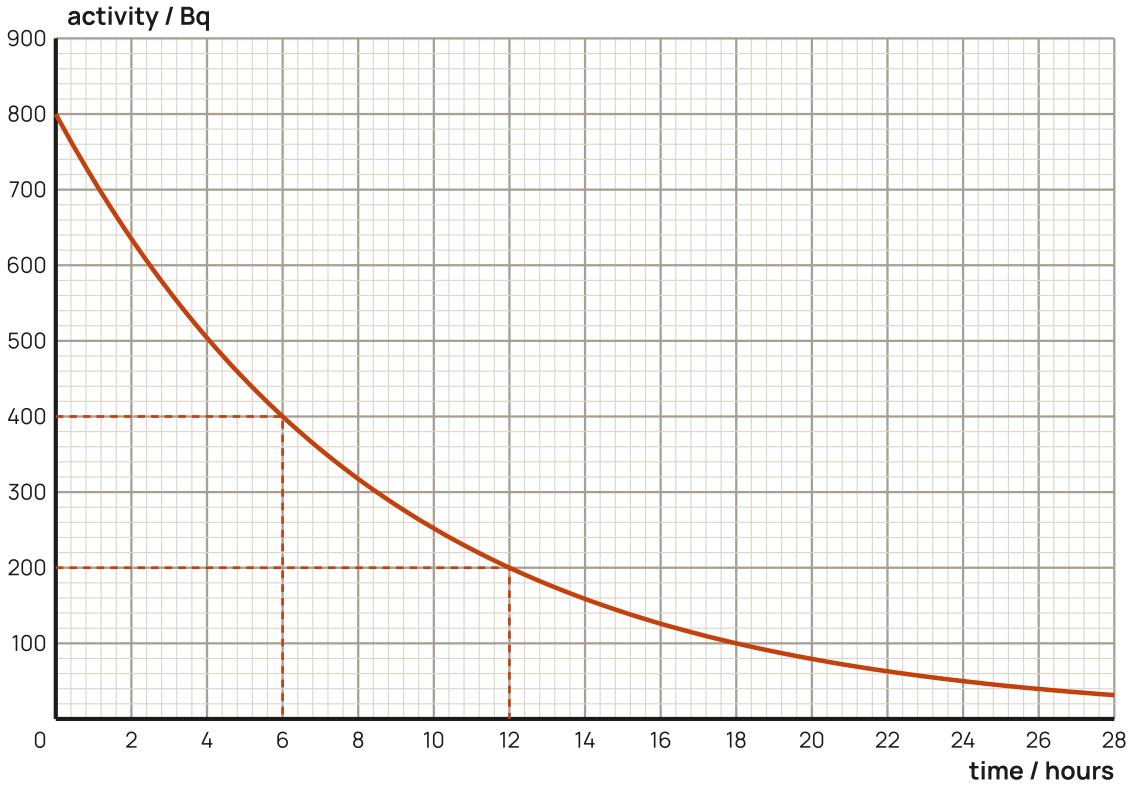
- the time it takes for the **number of unstable nuclei** in a sample to **halve**, or
- the time it takes for the **count rate** (or activity) from a sample to fall to **half** its initial level.

Each halving takes the same time. Iodine-131 has a half-life of about 8 days.

Finding the half-life from a graph

1. Read the **starting value**: here 800 Bq.
2. **Halve it**: 400 Bq.
3. Read **across** from 400 Bq to the curve, then **down** to the time axis: 6 hours.
4. **Check** with a second halving: 400 → 200 Bq also takes 6 hours.

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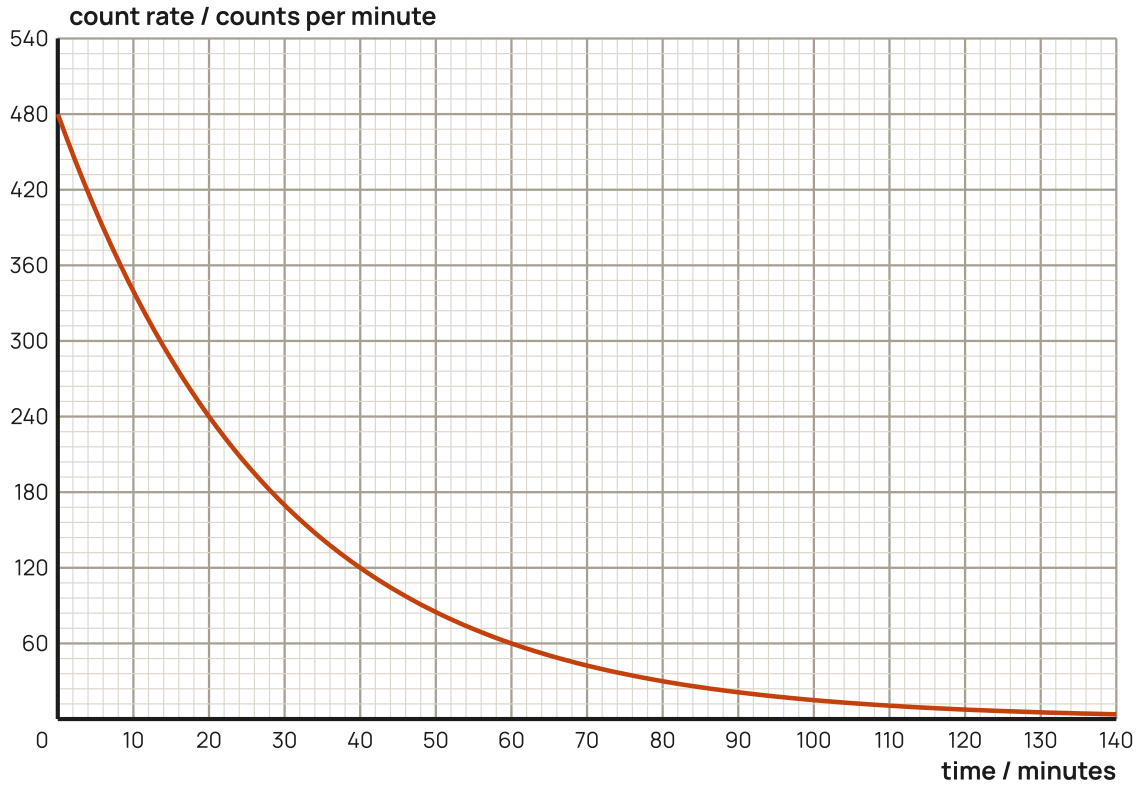
! Half-life = **6 hours**. Every halving takes the same time: 800 → 400 → 200 Bq at 6 and 12 hours.

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PRACTISE

Try it

1 The graph shows how the count rate from a sample changes. Use it to find:



(A) HALF-LIFE	
(B) COUNT RATE AT 40 MIN	
(C) TIME TO REACH 30	

2 The half-life of a sample is 5 days. Which statement is correct? Tick **one** box.

- ☐ All the nuclei will have decayed after 10 days.
- ☐ About half of the unstable nuclei will decay in the next 5 days.
- ☐ Each nucleus lasts exactly 5 days before it decays.
- ☐ After 5 days the sample is no longer radioactive.

3 SUPPORTED A sample has 1600 unstable nuclei. Its half-life is 3 hours. Complete the table.

Time / hours	0	3	6	9	12
Unstable nuclei left	1600				

4 Explain what is wrong with Leo's statement.



5 A student has a sample of only 100 nuclei. After one half-life, 44 are left, not 50. Explain why.

INVESTIGATE

Virtual lab: modelling half-life

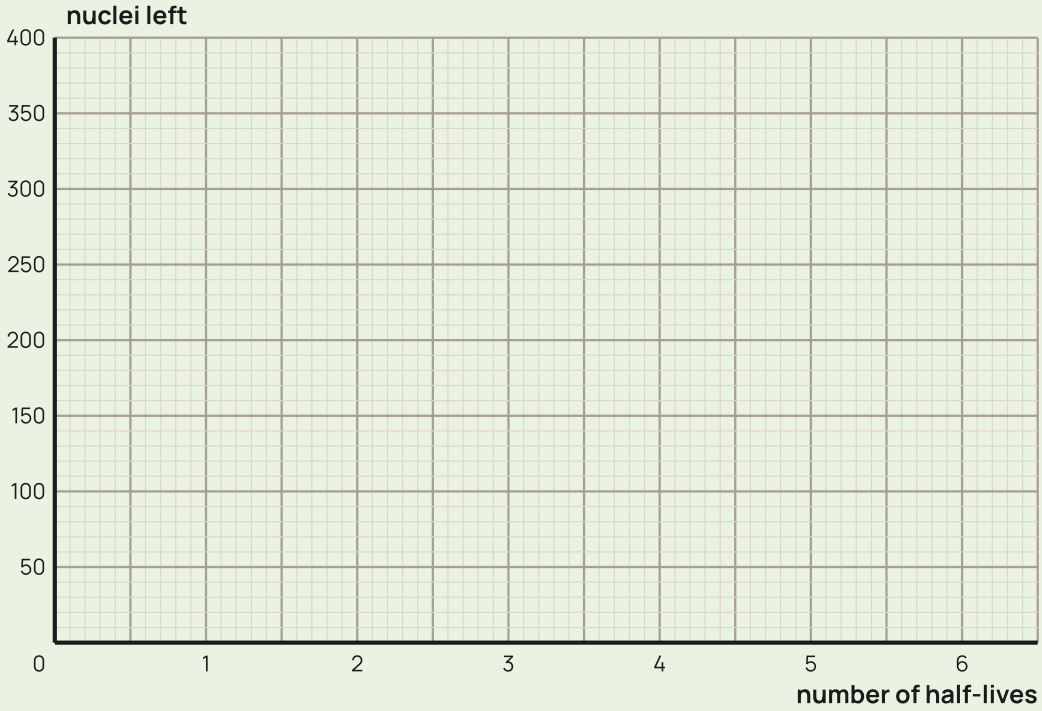
Nuclear Decay simulation, tab 04 (Half-life). No computer? Use 100 coins: each throw is one half-life; remove the heads.

- Open tab **04 Half-life**. Choose **Iodine-131** and **400 nuclei**.
- Before each step, write the number you expect. Then press **Run one half-life** and record the number left.
- Repeat until 5 half-lives have passed.

6 Record your results.

Half-lives	0	1	2	3	4	5
Expected number left	400					
Number left (your run)	400					

7 Plot your results. Draw a smooth curve of best fit.



8 Now run the simulation again with **100 nuclei**. How is the graph different? Explain why.

AQA-STYLE

Exam corner

Write in full sentences. Show every step of a calculation.

Get the marks: One definition is enough for 1 mark, but it must say 'half' and what is halving.

9 Define the half-life of a radioactive isotope. **[1 mark]**

10 Explain why the activity of a radioactive source decreases over time. **[2 marks]**

REVIEW LATER

Revise it

Use this box after the lesson and before tests.

SUMMARY

- Decay is random: we cannot predict which nucleus decays, or when.
- A large sample is predictable; small samples vary more.
- Half-life: the time for the number of unstable nuclei, or the count rate, to halve.
- From a graph: start value → halve it → across to the curve → down to the time.

COVER THE PAGE AND ANSWER

1 Define half-life.

2 What fraction is left after 2 half-lives?

3 Why are small samples less predictable?

4 What is the half-life of iodine-131?

I CAN…

- explain why decay is random but predictable
- define half-life in two ways
- find a half-life from a decay curve
- plot a decay curve from data

LESSON

09 Half-life calculations.

AQA Physics 4.4.2.3; Combined Science 6.4.2.3

BIG QUESTION

A hospital uses iodine-131, with a half-life of 8 days. How long before less than 1% of it is left, and why does that matter?

BY THE END YOU CAN

- count half-lives to find how much is left
- find a half-life from data
- find how long it takes to fall to a given value
- work out the net decline as a ratio

half-life | number of half-lives | initial activity | fraction remaining | net decline | ratio

SLIDES AND SIMULATIONS

Teaching deck, slide 36

Using half-life: from a graph, counting halvings, and as a ratio.

panphy.app/gcsephy/decks/atoms-and-radiation/#36

RETRIEVE

Do now

From earlier lessons. No notes!

1 Define half-life.

2 A count rate falls from 600 to 300 in 5 days. What is the half-life?

3 Why is radioactive decay described as random?

EXPLAIN

Counting the halvings

Three steps

1. Work out the **number of half-lives**: time ÷ half-life. Use the same units for both.
2. **Halve** the starting value that many times.
3. Give the answer **with a unit**.

WITH YOUR TEACHER SLIDE 36

A count rate falls from 400 to 200 counts per second in 8 days. What is the count rate after 24 days?

HALF-LIFE

HALF-LIVES

HALVE

ANSWER

YOUR TURN

A count rate falls from 960 to 480 counts per second in 6 hours. What is the count rate after 18 hours?

HALF-LIFE

HALF-LIVES

HALVE

ANSWER

WORKED EXAMPLE

A source has an activity of 640 Bq. Its half-life is 2 hours. Calculate its activity after 8 hours.

HALF-LIVES $8 \div 2 = 4$

HALVE $640 \rightarrow 320 \rightarrow 160 \rightarrow 80 \rightarrow 40$

ANSWER **40 Bq**

YOUR TURN

A source has an activity of 3200 Bq. Its half-life is 5 days. Calculate its activity after 15 days.

HALF-LIVES

HALVE

ANSWER

Finding the half-life from data

WORKED EXAMPLE

The activity of a sample falls from 2400 Bq to 300 Bq in 36 minutes. Calculate the half-life.

HALVINGS 2400 → 1200 → 600 → 300: 3 halvings

HALF-LIFE $36 \div 3$

ANSWER **12 minutes**

YOUR TURN

A count rate falls from 960 to 60 counts per minute in 20 hours. Calculate the half-life.

HALVINGS

HALF-LIFE

ANSWER

Net decline as a ratio

WITH YOUR TEACHER SLIDE 36

What fraction of the unstable nuclei is left after 3 half-lives? What is the net decline, as a ratio of the starting amount?

FRACTION LEFT

NET DECLINE

ANSWER

YOUR TURN

What fraction of the unstable nuclei is left after 5 half-lives? What is the net decline, as a ratio of the starting amount?

FRACTION LEFT

NET DECLINE

ANSWER

After n half-lives, the fraction left is $(½)^n$. The **net decline** is how much the activity has fallen. As a ratio of the initial activity, it is $1 - (½)^n$.

Example: 200 Bq falls to 50 Bq in 2 half-lives. Net decline = 200 − 50 = 150 Bq. As a ratio of the initial activity: 150 : 200 = **3 : 4**, or ¾.

Half-lives	Fraction left	Net decline
1	½	½
2	¼	¾
3	⅛	⅞
4	1/16	15/16

PRACTISE

Try it

1 SUPPORTED A source has an activity of 1200 Bq and a half-life of 6 hours. Calculate its activity after 24 hours.

HALF-LIVES	
HALVE	
ANSWER	unit:

2 ON YOUR OWN How long does it take for the activity of a source to fall from 1000 Bq to 125 Bq, if its half-life is 7 hours?

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3 Use the data to find the half-life. (Hint: find when the count rate is half of 800.)

Time / min	0	10	20	30	40
Count rate / counts per minute	800	566	400	283	200

4 Explain why Mia is wrong. What fraction is really left after 20 minutes?

Mia: The half-life is 10 minutes, so the activity will be zero after 20 minutes.

5 A source has an initial activity of 1200 Bq. Calculate its activity after 3 half-lives, and the net decline as a ratio of the initial activity.

6 CHALLENGE Iodine-131 has a half-life of 8 days. (a) What fraction is left after 32 days? (b) How many whole half-lives until less than 1% is left? How many days is that?

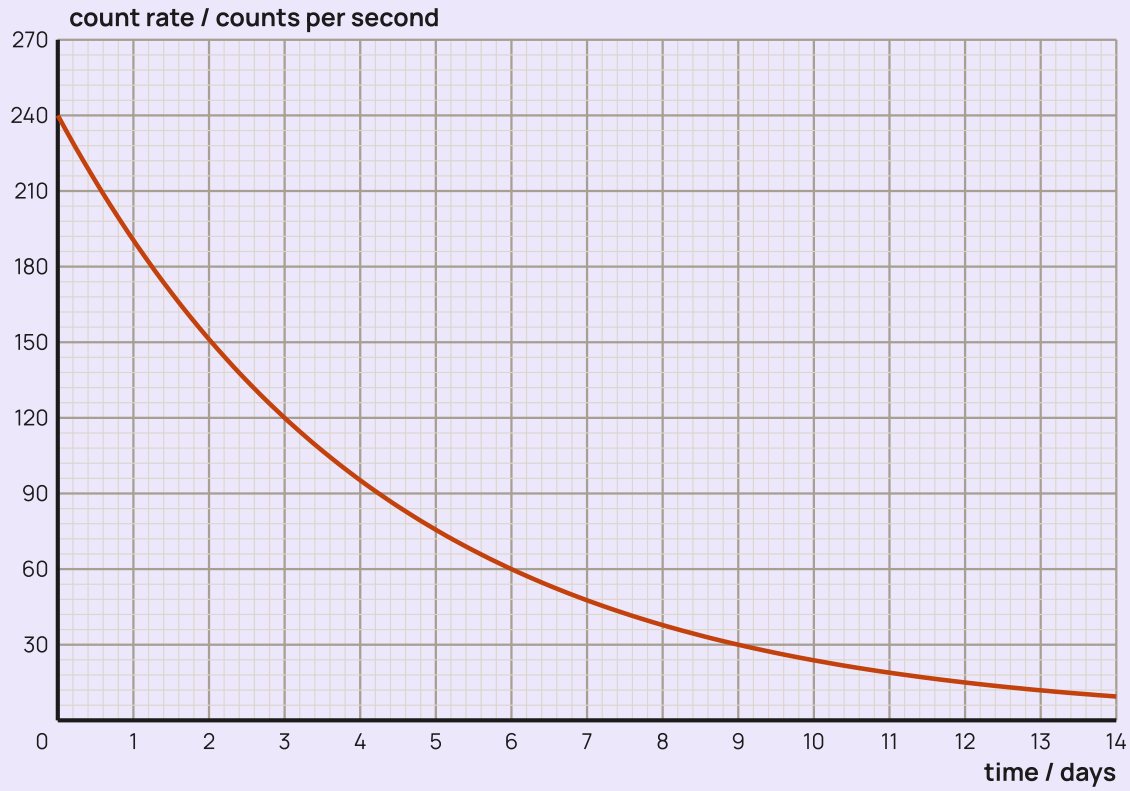
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Get the marks: Show the halvings, not just the answer. Units: keep kBq as kBq.

7 Cobalt-60 has a half-life of 5.3 years. A source has an activity of 400 kBq. Calculate its activity after 15.9 years. **[2 marks]**

8 Calculate the net decline in the activity of the cobalt-60 source after 15.9 years, as a ratio of its initial activity. **[2 marks]**

9 The graph shows the count rate from a sample. Determine the half-life. Show on the graph how you found it. **[2 marks]**



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REVIEW LATER

Revise it

Use this box after the lesson and before tests.

SUMMARY

- Number of half-lives = time ÷ half-life (same units).
- Halve the starting value once for each half-life.
- Half-life from data: count the halvings, then divide the time by that number.
- After n half-lives, $(½)^n$ is left.
- HT: net decline = initial − final; as a ratio of the initial, $1 - (½)^n$.

COVER THE PAGE AND ANSWER

1 What fraction is left after 4 half-lives?

2 800 Bq falls to 100 Bq. How many half-lives?

3 Half-life 4 h. How long for 640 Bq to fall to 80 Bq?

4 HT: net decline after 2 half-lives, as a ratio?

I CAN…

- calculate how much is left after a number of half-lives
- find a half-life from data or a graph
- find how long a source takes to fall to a value
- express the net decline as a ratio

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LESSON

10 Contamination and irradiation.

AQA Physics 4.4.2.4; Combined Science 6.4.2.4

BIG QUESTION

Supermarkets can sell food treated with gamma rays to kill bacteria. Does eating it make you radioactive?

BY THE END YOU CAN

- explain the difference between irradiation and contamination
- compare the hazards of alpha, beta and gamma inside and outside the body
- describe precautions for working with radioactive sources
- explain why studies of radiation effects are published and peer reviewed

irradiation | contamination | hazard | precaution | dosimeter | peer review

SLIDES AND SIMULATIONS

Teaching deck, slides 38–41

Contamination and irradiation, the quick check and the key words.

panphy.app/gcsephy/decks/atoms-and-radiation/#38

RETRIEVE

Do now

From earlier lessons. No notes!

1 A source has a half-life of 3 hours. What fraction is left after 9 hours?

2 Which radiation is the most penetrating?

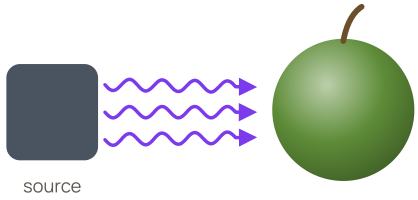
3 What stops beta radiation?

EXPLAIN

Two different things

Irradiation and contamination

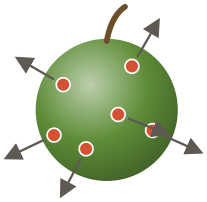
IRRADIATION



exposed to radiation

does NOT become radioactive

CONTAMINATION



radioactive atoms on or in it

keeps emitting until removed

	Irradiation	Contamination
What is it?	exposing an object to nuclear radiation from outside	unwanted radioactive atoms **on or in** an object
Does the object become radioactive?	**no**	it contains radioactive atoms, so it gives out radiation
When does the exposure stop?	when the source is removed or shielded	only when the radioactive atoms are removed (or have decayed)
Example	gamma rays used to sterilise food or medical equipment	breathing in radon gas; radioactive dust on skin

! An irradiated object does **not** become radioactive. A contaminated object keeps giving out radiation until the radioactive atoms are removed.

How big is the hazard?

Ionising radiation can damage cells and the DNA inside them: cells may die or start to divide out of control (cancer). The hazard depends on where the source is and which radiation it gives out.

Radiation	Source outside the body	Source inside the body
Alpha	low hazard: stopped by the outer layer of skin	**most hazardous**: strongly ionising, all its energy goes into nearby cells
Beta	can pass through skin to tissues below	hazardous: moderately ionising
Gamma	**most hazardous** outside: reaches organs inside	least hazardous inside: much of it passes out of the body

Staying safe

PRECAUTIONS WITH RADIOACTIVE SOURCES

- Handle sources with **tongs** or wear **gloves**: keeps the source away from the skin and stops contamination.
- Keep your **distance** and point sources away from people.
- Keep **exposure time** as short as possible.
- Store sources in **lead-lined boxes**, which absorb the radiation.
- Workers wear a **dosimeter badge** to measure how much radiation they receive.

Science you can trust

In the 1920s, workers painting watch dials with glowing radium paint licked their brushes to make a fine point. Many became seriously ill. Published studies of their health helped show how dangerous radioactive contamination is, and safety rules changed. It is important that findings about the effects of radiation on people are **published** and shared, so other scientists can check them by **peer review**.

PRACTISE

Try it

1 Is each example **irradiation** or **contamination**? Circle one for each.

A
Surgical instruments are sterilised with gamma rays.

irradiation

contamination

B
A person breathes in radon gas.

irradiation

contamination

C
Radioactive dust sticks to a worker's gloves.

irradiation

contamination

D
Strawberries are treated with gamma rays to keep them fresh.

irradiation

contamination

2 Explain why Leo is wrong.



3 SUPPORTED Explain how each precaution reduces the risk.

Precaution	How it reduces the risk
Using tongs	
Storing the source in a lead-lined box	
Keeping exposure time short	
Wearing a dosimeter badge	

4 Explain why an alpha source is the least hazardous outside the body but the most hazardous inside it.

5 Why is it important that studies of the effects of radiation on people are published? Tick **one** box.

- [ ] So that other scientists can check the findings (peer review).
- [ ] So that the scientists become famous.
- [ ] So that the results can never be changed.
- [ ] So that radiation becomes safe.

6 CHALLENGE Radon-222 is a gas given off by some rocks, and it emits alpha particles. Explain why radon in homes is a health risk, even though alpha cannot pass through skin.

CHECKPOINT

Quick check

Slide 40, Lessons 5–10. Answer each question, then check it as your teacher reveals the answer.

1
Complete: $^{210}_{84}\mathrm{Po} \rightarrow {}^{206}_{82}\mathrm{Pb} + ?$

2
In beta decay, how does the atomic number change?

3
Which radiation is the most ionising?

4
What stops beta radiation?

5
A source has a half-life of 6 hours. What fraction is left after 18 hours?

6
Does irradiated food become radioactive?

AQA-STYLE

Exam corner

Write in full sentences. Show every step of a calculation.

Get the marks: Define both words, and make the difference clear: outside source vs radioactive atoms on or in the object.

7 Explain the difference between irradiation and contamination. **[2 marks]**

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8 A technician uses a sealed gamma source. Describe two precautions the technician should take, and explain how each one reduces the risk. **[4 marks]**

REVIEW LATER

Revise it

Use this box after the lesson and before tests.

SUMMARY

- Irradiation: exposed to radiation; does not become radioactive.
- Contamination: radioactive atoms on or in an object; it keeps emitting until they are removed.
- Outside the body, gamma (and beta) are most hazardous; inside, alpha is most hazardous.
- Precautions: tongs, distance, short time, lead shielding, dosimeter.
- Findings about radiation's effects are published and peer reviewed.

COVER THE PAGE AND ANSWER

1 Does irradiated food become radioactive?

2 What is contamination?

3 Which radiation is most hazardous inside the body?

4 Why are results about radiation's effects published?

I CAN…

- explain the difference between irradiation and contamination
- compare the hazards inside and outside the body
- describe precautions and how each reduces the risk
- explain why findings are published and peer reviewed

REFERENCE

Toolkit.

Learn these: exam questions expect you to recall them without a data sheet.

The three particles

Lesson 1

Particle	Charge	Mass	Where
Proton	+1	1	nucleus
Neutron	0	1	nucleus
Electron	−1	very small	energy levels

Reading the symbol

Lesson 1

$^{A}_{Z}X$ e.g. $^{23}_{11}Na$

- Z, atomic number = protons
- A, mass number = protons + neutrons
- neutrons = A − Z; in an atom, electrons = Z
- Atom radius ≈ 1×10^{-10} m; nucleus radius < 1/10 000 of that

Isotopes and ions

Lesson 2

- **Isotopes**: same number of protons, different numbers of neutrons.
- **Positive ion**: an atom that has lost one or more outer electrons. The nucleus is unchanged.

Alpha scattering

Lesson 3

Observation	Conclusion
most straight through	mostly empty space
some deflected	positive centre
a very few bounced back	mass in a tiny centre

Models and energy levels

Lesson 4

- 1897 electron → 1904 plum pudding → 1911 nuclear → 1913 Bohr → 1932 neutron (Chadwick)
- **Absorb** EM radiation → higher level, further out.
- **Emit** EM radiation → lower level, closer in.

Nuclear radiation

Lesson 5

- Unstable nuclei decay at **random** to become more stable.
- **Activity**: decays per second, in becquerel (Bq). **Count rate**: decays recorded per second by a detector.
- α: 2p + 2n; β: fast electron, a neutron → proton; γ: EM radiation; n: a neutron

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Nuclear equations

Lesson 6

Emitted	Symbol	Mass no.	Atomic no.
alpha	$^{4}_{2}\mathrm{He}$	− 4	− 2
beta	$^{0}_{-1}\mathrm{e}$	no change	+ 1
gamma	γ	no change	no change
neutron	$^{1}_{0}\mathrm{n}$	− 1	no change

Top numbers balance; bottom numbers balance.

$$^{226}_{88}\mathrm{Ra} \rightarrow {}^{222}_{86}\mathrm{Rn} + {}^{4}_{2}\mathrm{He}$$

$$^{14}_{6}\mathrm{C} \rightarrow {}^{14}_{7}\mathrm{N} + {}^{0}_{-1}\mathrm{e}$$

Properties of radiation

Lesson 7

	Alpha	Beta	Gamma
Ionising	strongly	moderately	weakly
Range in air	a few cm	about 1 m	very far
Stopped by	paper, skin	a few mm of aluminium	reduced by thick lead or concrete
Typical use	smoke alarm	thickness gauge	medical tracer

More ionising → less penetrating.

Half-life from a graph

Lesson 8

1. Read the start value.
2. Halve it.
3. Across to the curve, down to the time axis.
4. Check with a second halving.

Half-life calculations

Lesson 9

- Number of half-lives = time ÷ half-life.
- After n half-lives, $(½)^n$ is left: ½, ¼, ⅛, 1/16…
- Net decline = initial − final; as a ratio of the initial, $1 - (½)^n$: ½, ¾, ⅞…

Contamination and irradiation

Lesson 10

- **Irradiation**: exposed to radiation; does not become radioactive.
- **Contamination**: radioactive atoms on or in an object.
- Outside the body γ is most hazardous; inside, α is.
- Tongs, distance, short time, lead, dosimeter.

REFERENCE

Key words from this booklet.

Learn the meaning and spelling. In exams, precise words earn marks: say “nucleus”, not “middle”, and “unstable nucleus”, not “unstable atom”.

Contamination: Unwanted radioactive atoms on or in an object.

Dosimeter: A badge that measures how much radiation a worker receives.

Half-life: The time for the number of unstable nuclei in a sample, or the count rate, to halve.

Irradiation: Exposing an object to nuclear radiation. The object does not become radioactive.

Net decline: How much the activity has fallen; often given as a ratio of the initial activity.

Random: Impossible to predict for one nucleus: which one decays next, or when.

CHECK YOURSELF

Quick answers.

Answers to the “Do now” and “Cover the page” questions, so you can check yourself when revising. Answers to the longer questions are in the answer edition.

LESSON 8: HALF-LIFE AND RANDOM DECAY

DO NOW

1. Alpha.
2. The number of decays a detector records each second.
3. Becquerel (Bq).

COVER THE PAGE

1. The time for the number of unstable nuclei (or the count rate) to halve.
2. A quarter (1/4).
3. Decay is random, so the numbers vary more around the expected value.
4. About 8 days.

LESSON 9: HALF-LIFE CALCULATIONS

DO NOW

1. The time for the number of unstable nuclei (or the count rate) to halve.
2. 5 days.
3. You cannot predict which nucleus will decay, or when.

COVER THE PAGE

1. 1/16
2. 3
3. 3 half-lives = 12 hours.
4. ¾ (3 : 4)

LESSON 10: CONTAMINATION AND IRRADIATION

DO NOW

1. ⅛ (3 half-lives).
2. Gamma.
3. A few mm of aluminium.

COVER THE PAGE

1. No.
2. Unwanted radioactive atoms on or in an object.
3. Alpha.
4. So other scientists can check them (peer review).